

Decision-making guide

Portfolio Management

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1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes introduction, market, portfolio allocation across three phases, Decision Checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

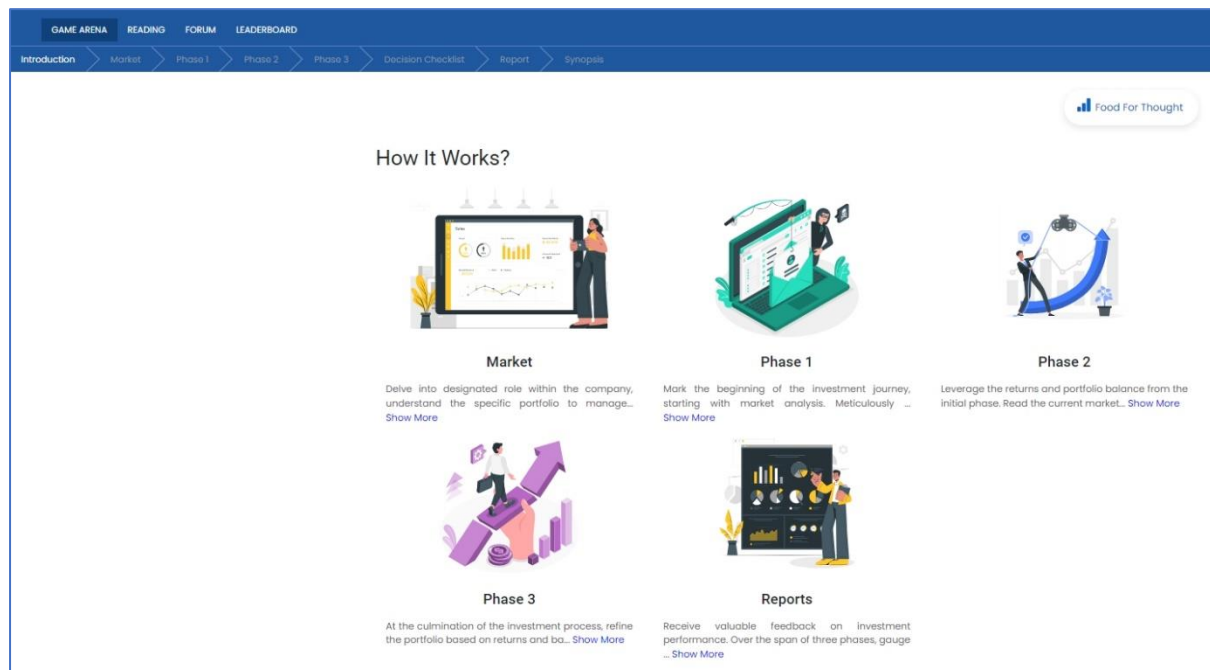


Figure 1 The Game arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the market, ongoing trends, obstacles, and the diverse array of decisions required.

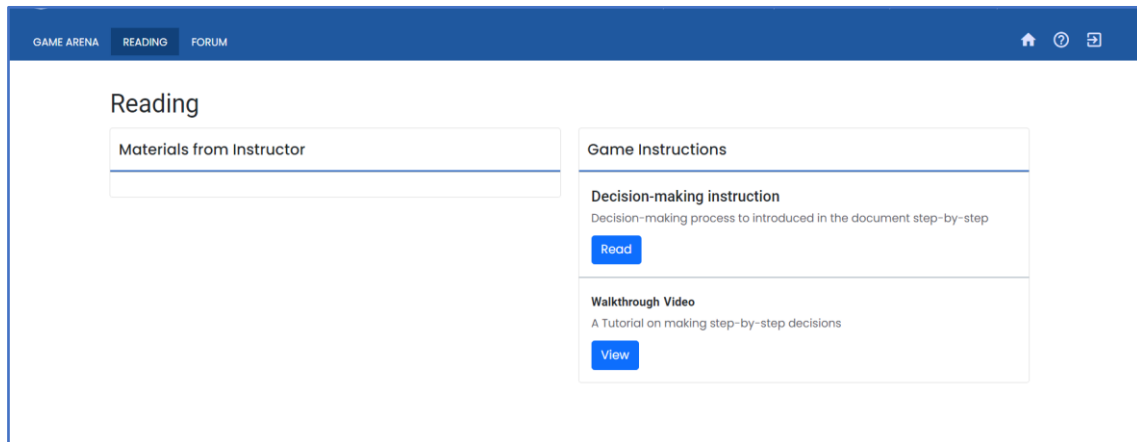


Figure 2 Reading Section in the platform

1.3 Forum

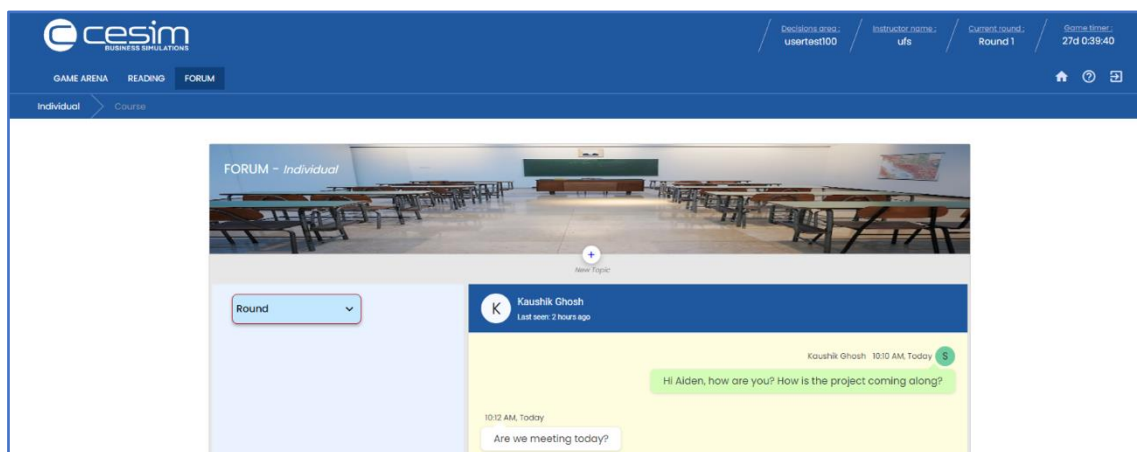


Figure 3 Forum Section in the platform

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

Keiser Capital Partners, founded in 1996, has grown into a diversified financial services firm with over 500 employees, headquartered in Mumbai and operating in 50+ cities. Serving a broad client base, Keiser Capital prioritizes a customer-first attitude, ethical practices, professionalism, research-based investing, and advanced technology. The participant is tasked with managing a new client's portfolio for the next 3 years, aiming for substantial returns above market benchmarks while maintaining lower unsystematic risk. The diversified portfolio may include stocks, commodities, bonds, funds, and cash equivalents. Success will be measured by returns and risk relative to the market, leveraging Keiser's research and ensuring client satisfaction.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

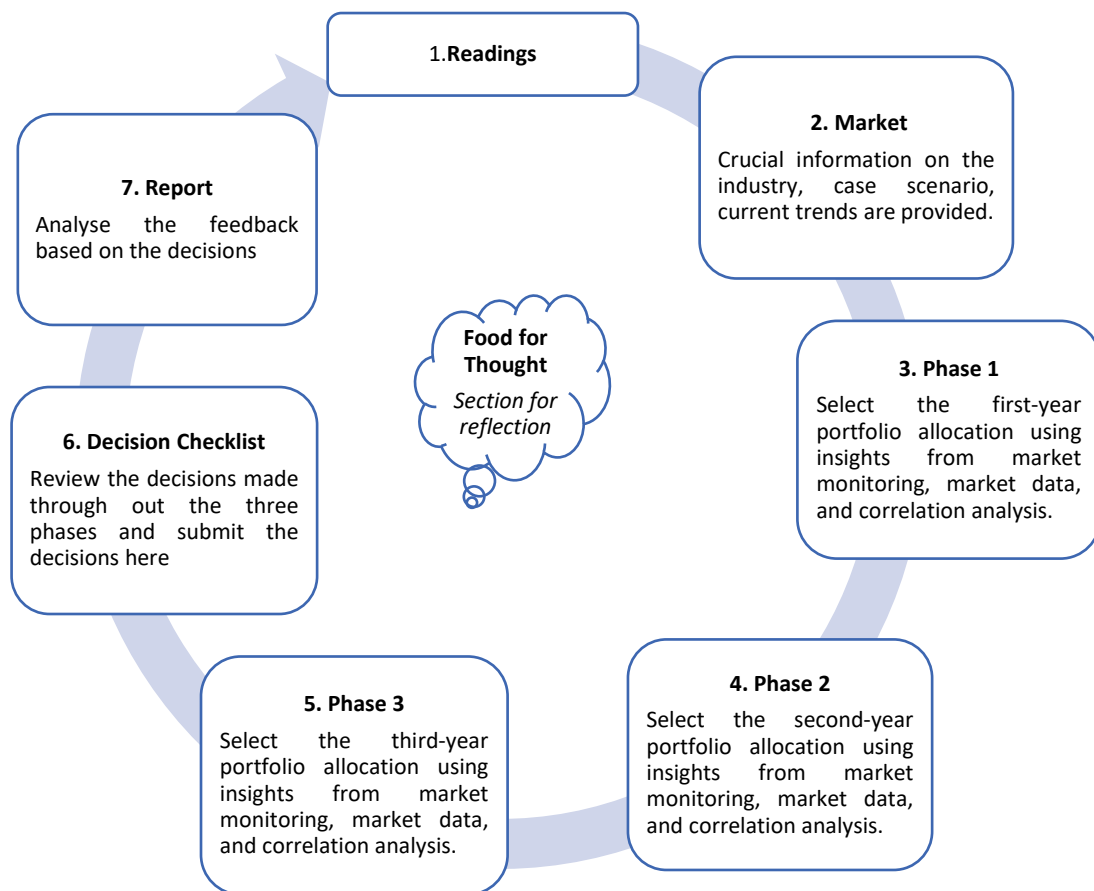


Figure 4 The sequence of decision making

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

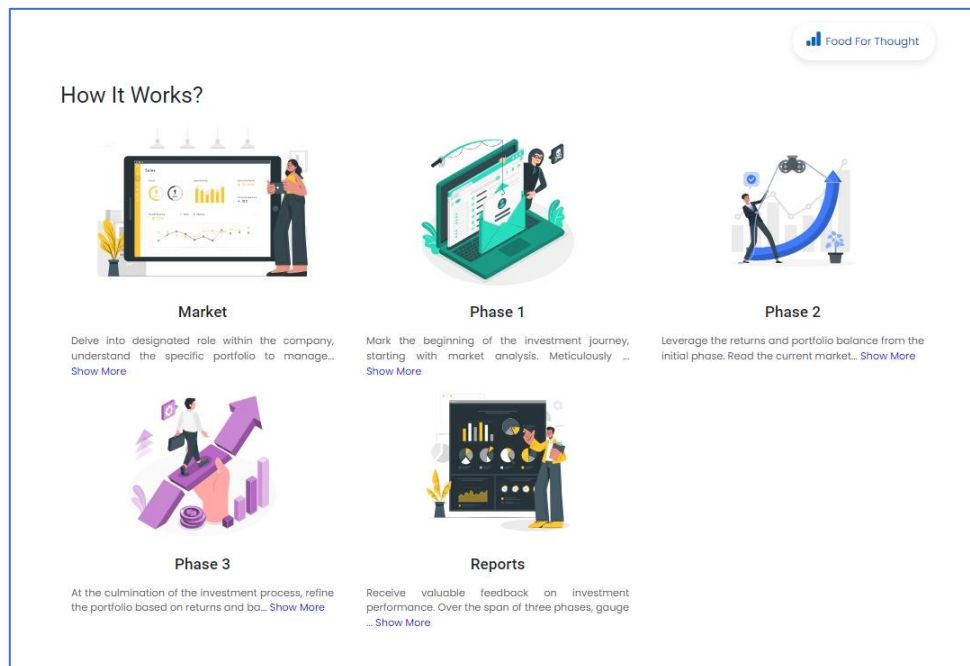


Figure 5 The Introduction Tab

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.

2 / 8
✕

While increasing allocation to high-risk securities might offer potential for higher returns, it could also amplify losses during market downturns. Consider diversifying to mitigate risks.

With an optimistic economic outlook projecting strong growth in specific sectors, how might you adjust your portfolio allocation?

- ☐ Increase allocation to sectors expected to experience robust growth.
- ☒ Diversify across sectors to mitigate risks associated with sector-specific fluctuations.
- ☐ Reduce exposure to sectors with uncertain growth prospects to minimize potential losses.



Figure 6 The food for thought Section in the simulation

2.3 Market and Parameters

Prior to commencing the decision-making process, it is essential to carefully examine the "Market" section. This section offers vital information on case scenario, industry overview and the key parameters to consider when making decisions in each tab.

Market
Phase 1
Phase 2
Phase 3
Decision Checklist
Report
Synopsis



Keiser Capital Partners was founded in 1996 as a small sub-broking unit, with just two people running the show. With a focus on a customer-first attitude, ethical and transparent business practices, respect for professionalism, research-based value investing, and implementation of cutting-edge technology has enabled us to blossom into a 500+ member team.

Today we are a well-diversified financial services firm offering a range of financial products and services such as private wealth, retail broking and distribution, institutional broking, asset management, investment banking, private equity, commodity broking, currency broking, and home finance. Our diversified client base includes retail customers (including High Net worth Individuals), mutual funds, foreign institutional investors, financial institutions, and corporate clients. We are headquartered in Mumbai and as of date, we have a network spread over 50 cities and towns comprising multiple business locations operated by our business partners, us, and 2,00,000+ customers.

Research is the solid foundation on which Keiser Capital's advice is based. Almost 10% of revenue is invested in securities research, and we hire and train the best resources to become our advisors. The firm has hired you with one objective to manage the portfolio of our newest client for the next 3 years. In this period, the portfolio has to produce substantial absolute returns over the market returns with lower unsystematic risk. The portfolio can comprise capital markets such as stock indices, commodities, bonds, funds, international stock indices and cash equivalents.

Parameters

	This Period
1 Phase is equal to	1 Year
Allocated Capital, k INR	10000
Security 1 - Indian Stocks	Replicate Nifty 50
Security 2 - Commodities	Replicate MCX
Security 3 - Hybrid Fund	60% equity + 17% debt + 15% low-risk debt
Security 4 - Corporate Bond	96% debt of which 26% in low risk securities
Security 5 - PSU Bond	96% debt of which 68% in low risk securities
Security 6 - US Stocks	Replicate Dow Jones
Security 7 - Cash Equivalents	100% low risk securities

Figure 7 The Market Tab

2.4 Market Monitor

The market monitor tab displays graphical data illustrating the historical returns of all portfolio components, along with their average annualized returns over the past nine years. This information helps in analysing performance trends and making informed investment decisions.

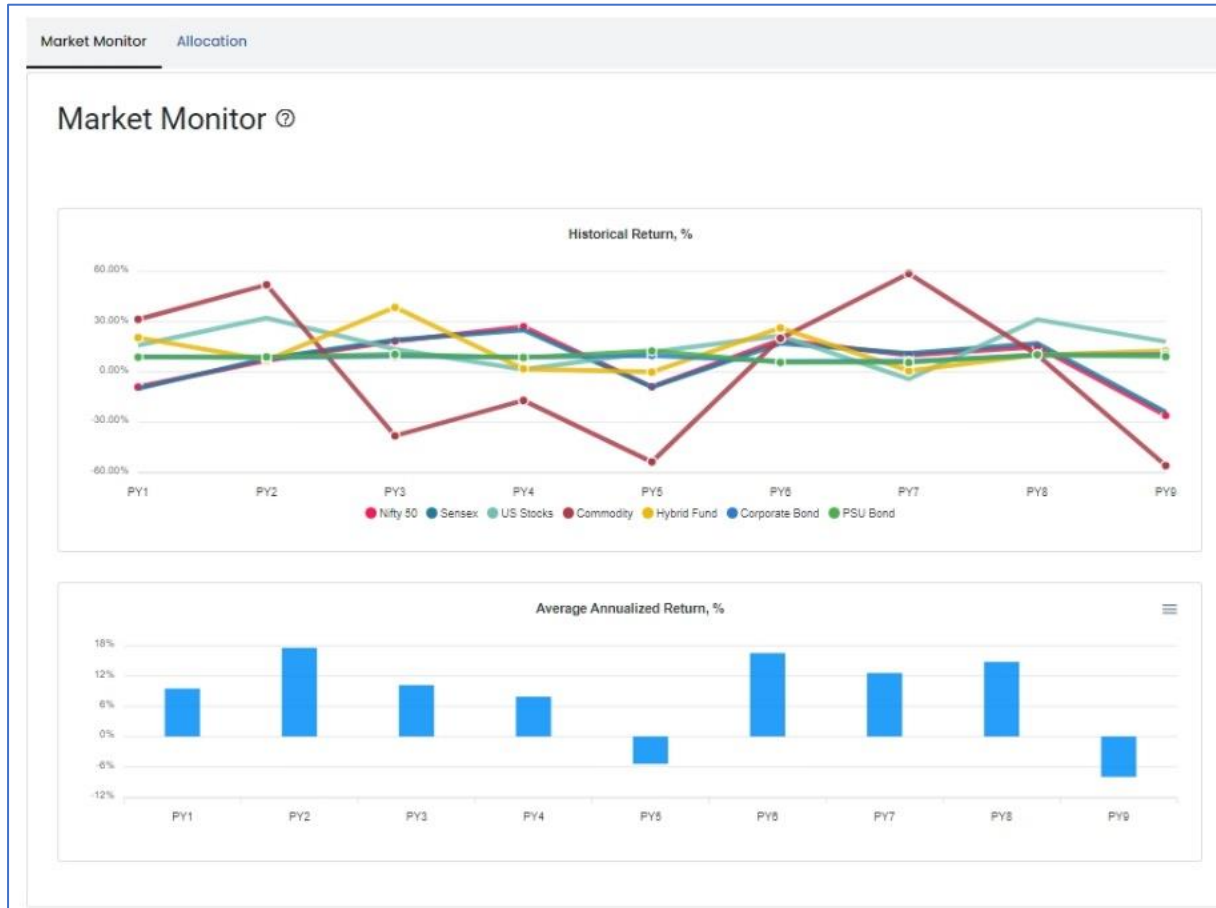


Figure 8 The Market Monitor Tab in the Platform

2.4 Allocation

Under the decision areas, the participants are tasked with determining the percentage allocation across various portfolios over the three years, based on the allocated funds and the parameters outlined in the market tab.

Phase 1

Analyse the market monitor data and the correlation table, which shows the relationships between various portfolios, to determine the best portfolio combinations. Allocate the initial 10,000 K INR for the first year accordingly. Once the allocation is made, proceed to phase 2.

Correlation ②

	Indian Stocks	Indian Commodity	Hybrid Fund	Corporate Bond	PSU Bond	US Stocks	Cash & Cash Equivalent
Expected Return	9%	8%	5%	6%	6%	17%	6%
Beta	0.39	0.19	-0.07	0	0	1.18	0.00007
Indian Stocks	1.00						
Indian Commodity	0.47	1.00					
Hybrid Fund	0.56	0.26	1.00				
Corporate Bond	0.67	0.36	1.00	1.00			
PSU Bond	0.78	0.22	0.98	0.99	1.00		
US Stocks	0.20	0.16	0.21	0.15	0.10	1.00	
Cash & Cash Equivalent	0.53	0.20	1.00	0.95	0.99	0.21	1.00

Figure 9 The correlation table available under the decision areas across three phases

Phase 2

Reassess the market monitor data to allocate the remaining funds for the second year.

Portfolio Allocation ②

Securities	Allocation	Value, k INR	
Indian Stocks	10 %	1022	Expected Return 6.7%
Indian Commodity	30 %	3065	Expected Risk 16.23%
Hybrid Fund	10 %	1022	Portfolio Beta 0.09
Corporate Bond	10 %	1022	
PSU Bond	0 %	0	
US Stocks	0 %	0	
Cash Equivalents	40 %	4087	

Phase 3

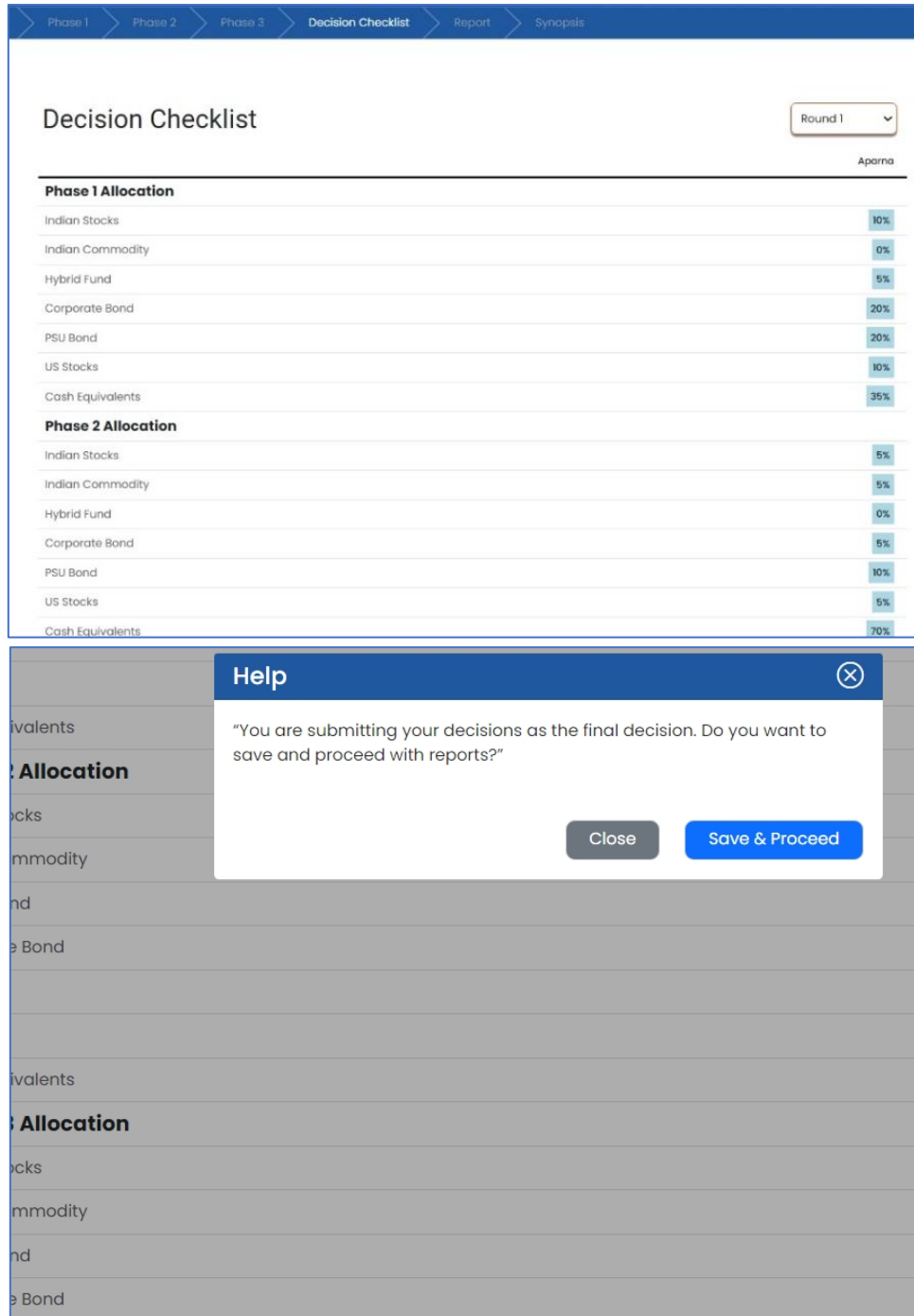
Figure 10 the portfolio allocation decision areas

Phase 3

Analyse the market monitor data to allocate the investment amounts for the third year. After making the allocation, proceed to the decision checklist.

2.5 Decision checklist

The Decision Checklist helps participants review their decisions. Once decisions are finalized, participants can submit them using the Submit option and proceed to save and proceed.



Decision Checklist

Round 1

Aparna

Phase 1 Allocation

Indian Stocks	10%
Indian Commodity	0%
Hybrid Fund	5%
Corporate Bond	20%
PSU Bond	20%
US Stocks	10%
Cash Equivalents	35%

Phase 2 Allocation

Indian Stocks	5%
Indian Commodity	5%
Hybrid Fund	0%
Corporate Bond	5%
PSU Bond	10%
US Stocks	5%
Cash Equivalents	70%

Help

"You are submitting your decisions as the final decision. Do you want to save and proceed with reports?"

Close Save & Proceed

Figure 11 The decision Checklist in the Platform

2.6 Report

- The report section provides the details on the participants' critical thinking abilities in each aspect. It demonstrates how their thought process influenced the outcomes, providing insights into the impact of their decision-making skills.

This section consists of feedback on the performance of the participant based on four metrics.

- **Rigour:** Consistency throughout the decision-making and recommendation.
- **Structuring:** Logical approach of gathering and analysing information
- **Synthesis:** Approach towards interpreting information and making informed decisions
- **Business Judgement:** Ability to evaluate facts, and consider risks and possible implications when making decisions.

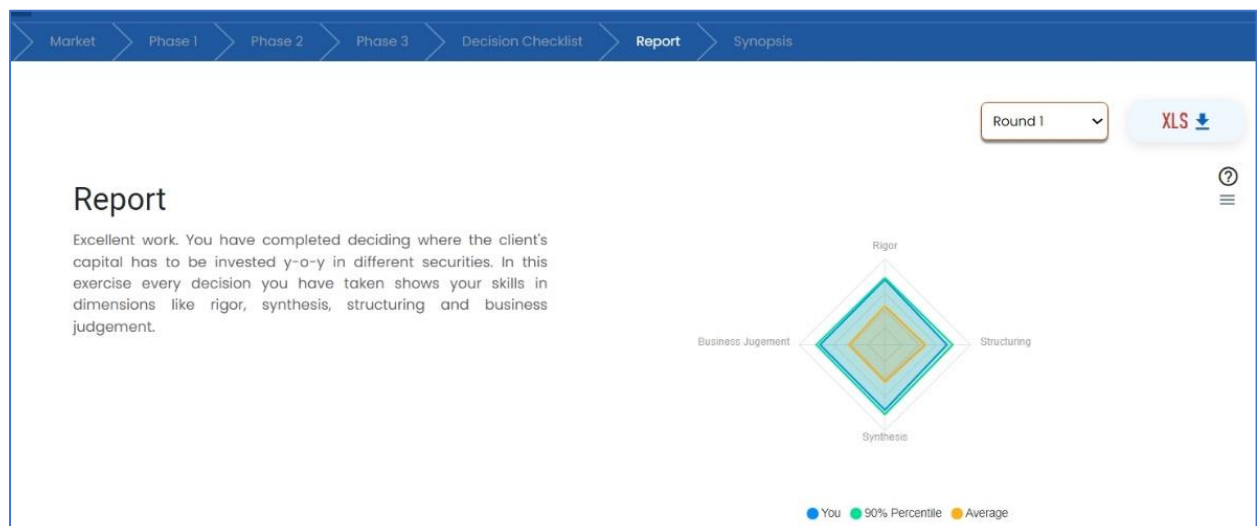


Figure 12 The report section

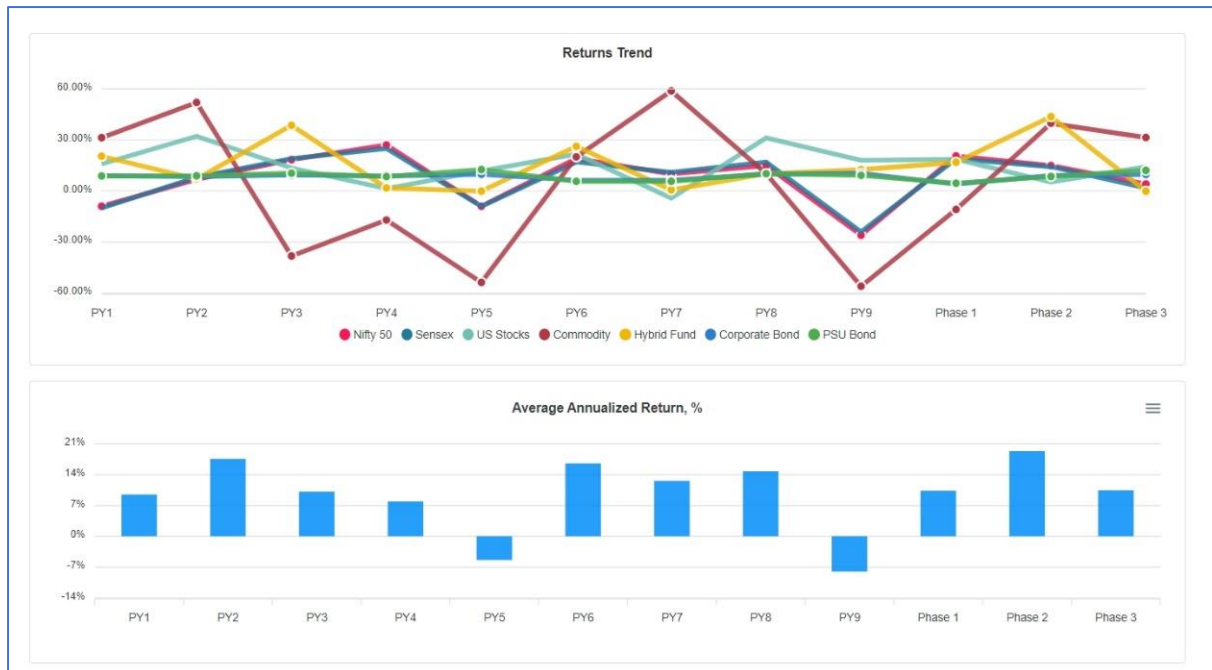


Figure 13 Historical Return Patterns and Average Annualized Returns Over Year

The phase measures section shows how well the participant has taken decision across all the three phases. The portfolio returns, risk and beta across three phases according to the allocation is given. The various ratios such as Treynor measure, Sharpe ratio, Jensen's alpha, CAPM and Fama Measure has also been calculated based on these. The capital invested across the securities are also provided.

Phase Measures				Capital Invested in Securities, k INR			
Parameter	Phase 1	Phase 2	Phase 3	Parameter	Phase 1	Phase 2	Phase 3
Portfolio Return	19.2%	9.22%	11.16%	Security 1 - Indian Stocks	1,000	596	1,302
Portfolio Risk	11%	26.68%	10.92%	Security 2 - Commodities	0	596	1,302
Portfolio Beta	0.48	0.63	0.57	Security 3 - Hybrid Fund	500	0	1,302
Treynor Measure	0.28	0.05	0.07	Security 4 - Corporate Bond	2,000	596	1,302
Sharpe Ratio	1.19	0.12	0.36	Security 5 - PSU Bond	2,000	1,192	3,906
Jensen's Alpha	7.91%	-2.38%	1.19%	Security 6 - US Stocks	1,000	596	1,302
CAPM	11.29%	11.61%	9.97%	Security 7 - Cash Equivalents	3,500	8,344	2,604
Fama Measure	4.12%	-3.54%	7.97%				

Figure 14 The report section in the Platform

- **Portfolio Return:** The total gain or loss generated by the portfolio, expressed as a percentage of the initial investment. A high return percentage indicates strong performance and substantial profit in the first year.
- **Portfolio Risk:** The standard deviation of the portfolio's returns, measuring the volatility or risk. A relatively low risk suggests the portfolio is stable and less prone to large fluctuations.
- **Portfolio Beta:** A measure of the portfolio's sensitivity to market movements. A Beta less than 1 indicates lower volatility than the market.
- **Treynor Measure:** A ratio that measures returns earned more than the risk-free rate per unit of market risk (Beta).
- **Sharpe Ratio:** A ratio that measures the excess return per unit of total risk (standard deviation).
- **Jensen's Alpha:** A measure of the portfolio's excess return over the expected return predicted by the Capital Asset Pricing Model (CAPM). An alpha of greater than 6 percentage indicates that the portfolio significantly outperformed the market expectations, reflecting the manager's skill in selecting investments.
- **CAPM:** The Capital Asset Pricing Model (CAPM) calculates the expected return of a portfolio based on its Beta (volatility relative to the market) and the overall market return. Typically, a "good" CAPM return would align closely with or slightly exceed the market return, considering the portfolio's risk profile.
- **Good Range:** The CAPM return is considered good if it aligns closely with or slightly exceeds the market return. This suggests that the portfolio is adequately compensated for the risk it bears. For example, if the market return is 10%, a CAPM return around 10% to 12% could be considered good.
- **Bad Range:** A CAPM return significantly below the market return indicates that the portfolio is not adequately compensating for its risk.
- For example, if the market return is 10%, a CAPM return significantly below 10% (e.g., 8% or lower) would be considered bad.

- **Fama Measure:** An indicator of the portfolio's performance based on factors such as market risk, size, value, and momentum. A positive Fama measure suggests good stock-picking skills and effective portfolio management.

Interpretation Table

Portfolio Measures	Good to Bad Range
Treynor Measure	- >0.25: Excellent - 0.15-0.25: Good - 0.10-0.15: Fair - <0.10: Poor
Sharpe Ratio	- >1.0: Excellent - 0.75-1.0: Good - 0.50-0.75: Fair - <0.50: Poor
Jensen's Alpha	- >6.0%: Excellent - 3.0%-6.0%: Good - 0.0%-3.0%: Fair - <0.0%: Poor
Fama Measure	- >3.0%: Excellent - 1.0%-3.0%: Good - 0.0%-1.0%: Fair - <0.0%: Poor

The combined measures are also provided that shows how participant has performed in total across three phases. The KPI (Key Performance Indicator) included the market return and the market risk.

Capital, k INR			
Parameter	Phase 1	Phase 2	Phase 3
Start	10,000	11,920	13,020
End	11,920	13,020	14,473

KPI	
Parameter	Output
Return over (+)/below (-) market	4.33%
Risk over (+)/below (-) market	-8.05%

Combined Measures	
Parameter	Output
Portfolio Return	39.43%
Portfolio Risk	30.76%
Portfolio Beta	0.6
Market	
Market Return	35.1%
Market Risk	38.81%

Figure 15 The report section showing the KPI of the simulation

2.7 Synopsis

The synopsis section provides a condensed overview of the key decision areas and outcomes, highlighting critical insights and strategic directions taken during the simulation. It offers participants a quick reference point to grasp the essence of their decisions and their impact on the key performance indicators thus aiding in holistic understanding and future planning.

Round 1

XLS

Synopsis

Decisions

Phase 1 Allocation	
Indian Stocks	10%
Indian Commodity	0%
Hybrid Fund	5%
Corporate Bond	20%
PSU Bond	20%
US Stocks	10%
Cash Equivalents	35%
Phase 2 Allocation	
Indian Stocks	5%
Indian Commodity	5%
Hybrid Fund	0%
Corporate Bond	5%
PSU Bond	10%
US Stocks	5%
Cash Equivalents	70%
Phase 3 Allocation	
Indian Stocks	10%
Indian Commodity	10%
Hybrid Fund	10%
Corporate Bond	10%
PSU Bond	30%
US Stocks	10%
Cash Equivalents	20%

Phase Measures

Parameter	Phase 1	Phase 2	Phase 3
Portfolio Return	19.20%	9.22%	11.8%
Portfolio Risk	11.00%	26.68%	10.92%
Portfolio Beta	47.51%	62.76%	57.44%
Treynor Measure	27.54%	5.37%	6.88%
Sharpe Ratio	118.93%	12.50%	38.06%
Jenson's Alpha	7.91%	-2.38%	1.19%
CAPM	11.29%	11.67%	9.97%
Fama Measure	4.12%	-3.54%	7.97%

Capital, k INR

Parameter	Phase 1	Phase 2	Phase 3
Start	10,000	11,920	13,020
End	11,920	13,020	14,473

Capital Invested in Securities, k INR

Parameter	Phase 1	Phase 2	Phase 3
Security 1 - Indian Stocks	1,000	598	1,302
Security 2 - Commodities	0	598	1,302
Security 3 - Hybrid Fund	500	0	1,302
Security 4 - Corporate Bond	2,000	598	1,302
Security 5 - PSU Bond	2,000	1,192	3,908
Security 6 - US Stocks	1,000	598	1,302
Security 7 - Cash Equivalents	3,500	8,344	2,604

KPI

Parameter	Output
Return over (+)/below (-) market	4.33%
Risk over (+)/below (-) market	-8.05%

Thinking Ability

Parameter	Output
Portfolio Management	88%

Combined Measures

Parameter	Output
Portfolio Return	39.43%
Portfolio Risk	30.78%
Portfolio Beta	0.8
Market	
Market Return	35.1%
Market Risk	38.87%

Figure 16 The Synopsis Tab